

LeLunar: Learning the Moon by Predicting One Instrument from Another

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Abstract

Orbital missions have mapped the Moon with many instruments, from multispectral images taken under changing sunlight to static maps of topography, roughness and composition. The products are co-registered but differ in resolution, units and physics, and labels are scarce. We propose LeLunar, a self-supervised model that learns from this stack by predicting, in representation space, what one instrument would record at a 60 km footprint given another instrument's view of it. Per-product stems give every product the same ground cell per token, from 60 m to 1 km per pixel. A cross-attention predictor, told the target product and the sun under which it was observed, predicts the token grid of a moving-average encoder. We hypothesize that this task forces the static maps to encode what shapes the appearance of the surface, and turns the predictor into a sun-conditioned renderer in latent space. We also argue that the design has a failure mode the loss hides: a position code, in which tokens depend on their place in the grid rather than on the terrain. We show why the loss, within-image rank and VICReg penalties should not see it, and why a covariance penalty and a variance hinge on the predictions should push toward it. We then propose tile-centered cross-sample monitors that the position, product and location channels cannot inflate. Finally, we lay out experiments that test each hypothesis, from controlled pretraining ablations with probe-based ground truth to crater detection, shape and albedo from shading, and a sun-swap test of the predictor. [TODO: headline results]

1. Introduction

Few places beyond Earth have been mapped as thoroughly as the Moon. The Lunar Reconnaissance Orbiter has returned camera images [71], laser altimetry [78], thermal radiometry [59] and radar [56] since 2009. Kaguya contributed multiband imagery and stereo topography [46, 57], and GRAIL a gravity field [99]. For almost any patch of ground one can look up a stack of co-registered maps. They record its brightness at five wavelengths under a given sun,

its height and slope, its roughness over tens of meters and its titanium or olivine content. What is scarce, as elsewhere in remote sensing, is labels.

Earth observation has answered that scarcity with foundation models pretrained on unlabeled multi-sensor archives [6, 24, 32, 44, 86, 92]. The Moon got its first ones in 2026 [31, 34, 64], and all three learn by reconstructing masked pixels or discrete tokens. We ask what a model learns if it must instead predict, in representation space, what one instrument would record at a place, given what another instrument saw there.

Lunar data suit the question for two reasons. The products are physically coupled. There is no atmosphere, so the shading in an image is set mostly by topography and the position of the sun, and albedo and roughness explain much of the rest. And every image comes with its illumination geometry, while the same terrain has been imaged many times under different suns. A predictor that is told the sun can therefore be asked what a footprint looks like under a particular light. To answer, the representation of the topography must carry whatever casts the shadows.

We build this as a joint-embedding predictive architecture (JEPA) [3, 13]. LeLunar encodes one product of a 60 km footprint with part of it hidden. A predictor, told the target product and the target's sun, then predicts the token grid that a slowly updated copy of the encoder produces for another product of the same footprint (Fig. 1). Each product, from 60 m to 1 km per pixel, gets a stem that reads its native pixels. As in OmniSat and AnySat [5, 6], a token then covers the same ground cell whatever its product, but we keep one token grid per product.

The same design is easy to break. Cross-modal latent prediction has an escape route that masked prediction within one image mostly lacks. Much of the target grid cannot be predicted from the source at the scale of a token, and the student and its target share cheap channels: token position, product identity and footprint location. A model can lower its loss by encoding those channels and letting the content of its targets fade. We call the end point a position code. It is a cousin of the positional collapse that CAPI [27] reported for latent masked image modeling, but with continuous targets

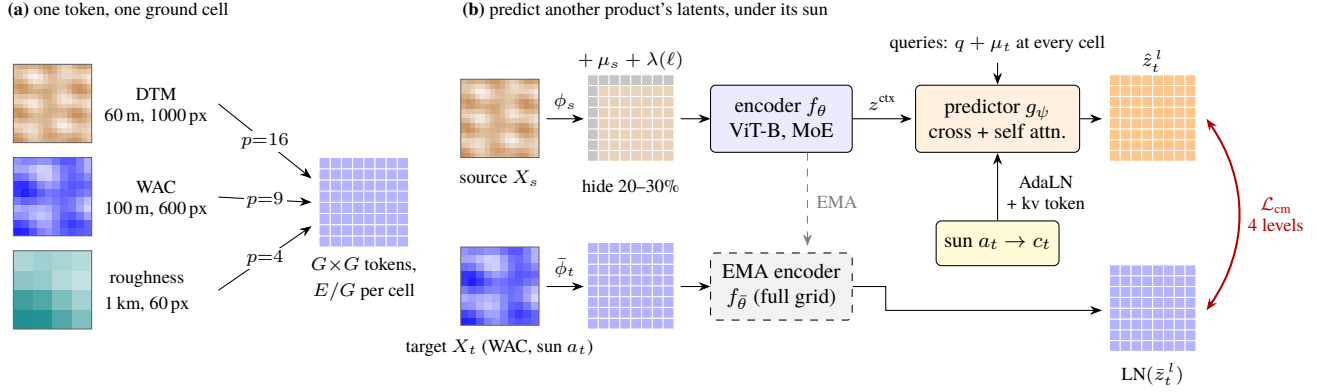


Figure 1. LeLunar in one picture. (a) Each product has its own stem, whose kernel equals its native pixels per token, so a 60 m DTM, a 100 m WAC image and a 1 km roughness map all become the same $G \times G$ grid of ground cells. (b) The encoder sees the source with 20–30% of its tokens hidden. The predictor’s queries carry the target product, and the target’s sun conditions every predictor layer through AdaLN and an extra key/value token. The target is the EMA encoder’s normalized output for the other product at four depths. The dense loss on visible source tokens (Eq. (5)) is not drawn.

and across modalities it should be harder to see. We argue that the loss, the effective rank within a tile and the VICReg penalties all improve as it forms, and that two common regularizers reward it.

This paper states these ideas as testable hypotheses and designs the experiments that decide them. Our contributions are:

- A cross-modal JEPa for co-registered planetary rasters. Each product keeps its own ground-aligned token grid at native resolution, from 60 m to 1 km per pixel. A predictor, told the target product and the sun of the target observation, predicts the dense token grid of another product (Sec. 3).
- Five hypotheses about what this objective learns and how it fails (Secs. 3.7 to 3.11). They include an analysis of position-code collapse: why loss, within-sample rank and VICReg are blind to it, and why the VICReg covariance penalty, a variance hinge on predictions and pooled SIGReg can push toward it.
- Tile-centered cross-sample monitors that the position, product and location channels cannot inflate (Sec. 3.10).
- An experimental protocol that tests each hypothesis (Sec. 4). It covers controlled pretraining ablations with probe-based ground truth, and downstream crater detection, shape and albedo from shading, and a sun-swap test.

[TODO: summarize results once the runs finish]

2. Related work

Latent prediction. I-JEPa [3] predicts the representations of masked image blocks from a visible block, with targets from an exponential moving average (EMA) of the encoder, a teacher also used by BYOL, DINO, iBOT and

data2vec [9, 20, 35, 58, 97]. V-JEPa and V-JEPa 2 do the same for video [4, 13], and V-JEPa 2.1 [55] adds supervision at four depths and a distance-weighted loss on visible tokens, both of which we adopt. Multimodal masked modeling [8, 53] predicts across modalities but reconstructs pixels or tokens. Predicting one modality’s latents from another’s dates back at least to audio-visual RAVeN [36]. In Earth observation the closest mechanism is Galileo’s [86]. Its predictor queries carry the target channel group and cross-attend to the other groups, with EMA targets, but inputs are re-sampled to 10 m and the loss is contrastive. AnySat [6] predicts fused multi-sensor cell embeddings. X-JEPa, CR-JEPa and Le MuMo JEPa [22, 25, 39] predict or regularize pooled embeddings, mostly for retrieval. To our knowledge no JEPa conditions its predictor on the illumination geometry of the target. The nearest precedents condition on the month (Galileo), the target view’s projection [63] or a caption [41].

Collapse and its diagnosis. Joint-embedding methods can collapse completely or into a low-dimensional subspace [40, 45, 84]. VICReg [12] and Barlow Twins [96] decorrelate features across a batch. C-JEPa [54] adds VICReg to I-JEPa. LeJEPa [10] replaces the teacher with a sliced test toward an isotropic Gaussian (SIGReg), whose gradients can vanish under collapse [90]. RankMe [33] and LiDAR [83] estimate quality from the embedding spectrum, and Latent MIM [89] monitors the cosine between pooled embeddings of different images. Objectives can also be met without content. Context prediction learned patch location from chromatic aberration [28], and JEPAs trained with VICReg can latch onto slow, trivially predictable features [79]. CAPI [27] reports positional collapse, in which

latent masked-image models predict the position of masked tokens instead of their content. It was found by inspecting feature maps and fixed by clustering targets per position. StoP [11] and SiamJEPa [93] make positions stochastic or shuffle the teacher. OlmoEarth [38] replaced EMA targets, which collapsed in its experiments, with a frozen random projection. Absolute position also leaks through zero padding [23, 43, 47], and self-supervised ViTs, including ones with rotary embeddings, carry input-independent positional components [61, 95]. We analyze positional collapse in a continuous-target, cross-modal JEPa, derive which monitors and regularizers should miss or reward it, and propose a monitor that can see it.

Resolution in Earth observation models. Most multi-sensor models resample their inputs to a common grid [32, 44, 86, 92]. The alternatives are GSD-scaled positional codes [30, 42, 68] and per-band or per-sensor patch sizes [42, 88]. OmniSat and AnySat [5, 6] cut all modalities into the same ground cells, and we follow them. They fuse a cell’s modalities into one token, however, which rules out predicting one product from another. Location encoders on the sphere [48, 52, 72, 87] provide a geographic prior. We use Slepian functions [77], as proposed for location encoding by Rao *et al.* [67].

The Moon and other planets. Lunar machine learning has mostly meant task-specific models. Examples are crater detection [75, 82, 94], rockfall and boulder mapping [7, 15, 16], imaging of permanently shadowed regions [17] and single-view topography [21, 73]. The NASA-IBM lunar foundation model [31] retrains TerraMind’s masked-token recipe on WAC and NAC imagery and topography from SomBench [60]. It takes acquisition geometry as encoder input, and static maps enter it as tile-averaged scalars. LunarFM [34] is a MultiMAE over 18 global channels on 15 km chips and names JEPa-style objectives as future work. Moonstone [64] is a modality-grouped MAE at 237 m. On Mars, MOMO [66] merges per-sensor MAEs, and Mars-Bench [65] standardizes evaluation. None of these models predicts latents across modalities. LeLunar keeps static maps as spatial inputs and targets at native resolution, and learns from repeat observations under different suns.

3. Method

We first describe the data and the model (Secs. 3.1 to 3.6), then state five hypotheses about what the objective learns and how it fails (Secs. 3.7 to 3.11). Section 4 tests them.

3.1. Co-registered lunar footprints

Our data are 60 km $\times$ 60 km footprints of the lunar surface, with every available orbital product cut to the same ground

Table 1. Products used in this paper. r : native ground sample distance. C : bands. p : native pixels per token at $G=64$ and $S_{\max}=1000$, before and after a kernel floor of 4 (Sec. 3.3). *The appearance depends on the sun, and each observation carries its illumination angles. The lower block joins in the 12-product runs.

Product	Source	r (m)	C	p
WAC visible*	LROC WAC [71]	100	5	9
WAC ultraviolet*	LROC WAC	500	2	2 $\rightarrow$ 4
Elevation	SLDEM2015 [14]	60	1	16
Slope; aspect	from elevation	60	1; 2	16
643 nm mosaic	WAC normalized [18]	100	1	9
Roughness	LOLA [49]	1000	1	1 $\rightarrow$ 4
Morphologic mosaic	WAC [80]	100	1	9
Normalized reflectance	WAC normalized [18]	500	7	2 $\rightarrow$ 4
MI reflectance	SELENE MI [50]	60	8	16
MI mineralogy	SELENE MI [50]	60	7	16
TiO ₂	WAC [74]	400	1	2 $\rightarrow$ 4

extent and map projection. A footprint appears once per Wide Angle Camera (WAC) observation, so its static maps are paired with images taken under different suns. The full set has 10,629 footprints and 476,635 rows. Table 1 lists the products, from 60 m per pixel (topography, Kaguya Multi-band Imager) to 1 km (LOLA roughness). Diviner bolometric temperature and GRAIL gravity are too coarse to tokenize and serve only as an auxiliary target (Sec. 3.6). Each WAC row stores the incidence, emission and phase angles and the sub-solar azimuth, latitude and longitude of the acquisition. Splits follow Lunar Transverse Mercator zones, so that no validation footprint overlaps a training one. We exclude polar footprints. **[TODO: non-polar split counts]**

Values outside gross physical bounds are marked missing, the rest are clipped to ranges set by instrument scientists, and heavy-tailed products are log-transformed. Rasters larger than $S_{\max}$ pixels per side are area-averaged down, ignoring missing pixels. Every band is standardized with training statistics, and missing pixels are then set to zero, the mean.

3.2. Task

For each row we shuffle the available products and take the first as the source s and the next $K=2$ as targets t_1, t_2 . The source is encoded with part of its token grid hidden. For each target the model predicts the full token grid that an EMA of the encoder produces from the target raster, which the student never sees. It is told only which product to predict and, for WAC targets, the sun of that observation. Since each footprint has several WAC rows, the same topography must be turned into images under different suns.

3.3. One token, one ground cell

The simple alternative is to resample all products to a shared $4G \times 4G$ canvas with 4×4 patches. At $G=64$ that shrinks the

1000 px topography raster four times and the 600 px WAC image 2.3 times, and blows the 60 px roughness map up four times. We instead align tokens to ground cells, as OmniSat and AnySat do [5, 6], and give every product m its own stem ϕ_m , whose kernel and stride equal its native pixels per token,

$$p_m = \max(k_{\min}, \text{round}(\min(E/r_m, S_{\max})/G)), \quad (1)$$

where $E=60$ km is the footprint extent. The raster is resized to Gp_m pixels per side, which changes it by a few percent unless the floor below applies, and ϕ_m maps it to a $G \times G$ grid of D -dimensional tokens. Every token is then one ground cell of E/G meters (940 m at $G=64$), whatever the product.

Kernel floor. At $G=64$, Eq. (1) with $k_{\min}=1$ gives LOLA roughness one pixel per token. Each token is then a scalar times a fixed vector, so all roughness tokens lie on a line in $\mathbb{R}^D$. Such a target has rank one by construction, is easy to fit and carries little. A floor $k_{\min}=4$ pre-upsamples coarse products so that each token sees a 4×4 patch.

Convolutional stems. A linear patch projection [29] must preserve edges and texture through one matrix, so we also use factorized convolutional stems [91]. p_m is factored into stride-3 and stride-2 stages ($9=3 \cdot 3$, $16=2^4$, $4=2^2$), each a convolution followed by GroupNorm and GELU, with channel width doubling toward $D/2$, then a 1×1 projection to D . The usual 3×3 kernel with padding 1 has a quiet flaw here, a case of the receptive-field offsets that stride and padding produce [1, 2]. It centers output o of a stride- s stage on input pixel so rather than on the center $so + (s-1)/2$ of its cell. The shifts add up across stages to

$$\delta_m = -\frac{p_m - 1}{2p_m} \text{ cells}, \quad (2)$$

which is -0.38 , -0.44 and -0.47 of a token for $p_m = 4, 9, 16$. Because the shift depends on p_m , tokens of different products refer to ground up to 0.09 cells apart. A kernel of size $s+2$ with padding 1 centers the output on its cell, overlaps its neighbours at every stride and keeps the exact $G \times G$ output. We use it in all stride-2 and stride-3 stages (Appendix A).

3.4. Encoder, location and experts

The encoder f_θ is a ViT-B [29] (12 blocks, $D=768$, 12 heads). Positions enter only through axial 2D rotary embeddings [37, 81] on queries and keys. The last four blocks replace the MLP with a sparse mixture of experts: 8 experts, top-2 routing, a router with a learned bias per product, and the V-MoE load-balancing loss [69]. A token enters the encoder as

$$e_{m,i} = \text{LN}(\phi_m(X_m)_i) + \mu_m + \lambda(\ell), \quad (3)$$

where μ_m is a learned product embedding and $\lambda(\ell)$ encodes the footprint center ℓ with 1536 Slepian functions [67, 77] (band limit 140) and an MLP. Only the tokens of the visible set V reach the encoder. V is one square block covering 70–80% of the grid, drawn per batch, so 20–30% of the source is hidden. The encoder returns layer-normalized features after blocks 3, 6, 9 and 12. A linear layer fuses them into the context $z^{\text{ctx}} \in \mathbb{R}^{|V| \times D}$.

The target branch is an EMA copy of the whole input pipeline: stems, product and location embeddings, and encoder. It reads the full, unmasked target raster and returns four layer-normalized feature grids $\tilde{z}_t^l \in \mathbb{R}^{G^2 \times D}$.

3.5. A predictor that is told the sun

The predictor g_ψ is a 12-layer decoder as wide as the encoder. I-JEPA keeps its predictor narrow [3], but translating between products is harder than filling in part of one image. Its G^2 queries are a learned vector plus the target product embedding, made distinct by rotary embeddings of their cells. Each layer cross-attends from the queries to the context, self-attends among the queries and applies an MLP. The illumination angles a_t are encoded as cosines and sines of incidence, emission, phase, sub-solar azimuth and longitude, plus sub-solar latitude and a validity bit, which is off for static targets. A small MLP turns this into a condition vector c_t that sets the scale and shift of every LayerNorm in the predictor (AdaLN-Zero [62], initialized to the identity). A projection of c_t is also appended to the cross-attention keys and values as one extra token. Four heads, one per supervision level, read the final query states.

3.6. Objective

With ρ the smooth- ℓ_1 loss and LN a LayerNorm without affine parameters, the cross-modal loss for target t is

$$\mathcal{L}_{\text{cm}} = \frac{1}{L} \sum_{l=1}^L \frac{1}{G^2 D} \sum_{i,d} \rho(\hat{z}_{t,i,d}^l - \text{LN}(\text{sg}[\tilde{z}_{t,i}^l]_d)), \quad (4)$$

with $L=4$ and sg the stop-gradient. We add a dense context loss in the spirit of V-JEPA 2.1 [55]. Each visible source token at every level is pulled toward the EMA encoding of the full, unmasked source at the same cell. The weight of the token depends on its distance d_i (in cells) to the nearest hidden cell. V-JEPA 2.1 weights by $1/\sqrt{d_i}$; we use an exponential that decays faster:

$$\mathcal{L}_{\text{ctx}} = \frac{1}{L} \sum_l \frac{1}{|V|} \sum_{i \in V} w_i \bar{\rho}(z_{s,i}^l, \text{LN}(\tilde{z}_{s,i}^l)), \quad (5)$$

where $\bar{\rho}$ averages ρ over features and $w_i \propto e^{-(d_i-1)/\tau}$ with $\tau=2$ cells, normalized to mean one. Tokens next to the hidden region, where the full view knows more than the masked one, carry most of the weight. The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{cm}} + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}} + w_v \mathcal{V} + w_c \mathcal{C} + \alpha \mathcal{L}_{\text{moe}} + \beta \mathcal{L}_{\text{cond}}, \quad (6)$$

where $\mathcal{V}$ and $\mathcal{C}$ are the VICReg variance hinge and covariance penalty [12]. Both are computed on the tokens of z^{ctx} and of the final-level prediction, flattened over the batch. $\mathcal{L}_{\text{cond}}$ is an ℓ_1 loss for a small head that regresses the footprint-mean Diviner temperature (24 local-time bins) and GRAIL gravity from $\lambda(\ell)$. We use $\lambda_{\text{ctx}}=0.5$, $w_v=1$, $w_c \in \{0.005, 0.01\}$, $\alpha=10^{-3}$ and $\beta=0.01$. The EMA decay is 0.996, optionally ramped to 0.9995 with a cosine schedule. Section 3.9 explains why we keep the regularizer weights small and why we expect two of the terms to do harm.

3.7. What the objective should learn

On an airless body the appearance of a footprint is close to a deterministic function of its topography, its albedo and the geometry of illumination and viewing. Shading follows the slope and aspect of each facet relative to the sun, shadows follow the relief along the sun’s azimuth, and albedo scales the result. A predictor that must produce the latents of a WAC image from those of a DTM and a sun can succeed only if the DTM representation carries slope- and aspect-like content for the sun to act on. The reverse task rewards features that do not change with the sun, because the DTM target is the same for every observation of a footprint.

H1. Cross-modal latent prediction under a specified sun makes the static-product representations carry the terrain properties that determine appearance, and makes the predictor a sun-conditioned renderer in latent space. Then frozen features should transfer better to terrain tasks than those of a same-product JEPA with the same data and compute, shadows decoded from predicted image latents should follow the sun they are given, and relief read from an image should not change between observations of a footprint.

3.8. A position-code attractor

Write a target token of product t at cell i of footprint x as

$$\bar{z}_t(i, x) = u(i) + m_t + l(\ell_x) + c_t(x, i), \quad (7)$$

a position term, a product term, a location term and the rest, which we call content. The predictor gets u , m_t and l for free: rotary embeddings give it the query cell, the query carries the target product, and the location embedding is added to every token on both paths. Only c_t requires reading the source, and for many pairs it is barely predictable at the token level (a 100 m image says little about kilometre-scale roughness). There the student can lower the loss only by shrinking c_t in its own features, and the EMA target inherits what the student does. Predicting $u+m_t+l$ and letting c_t fade is thus a descent direction. In a same-image JEPA the pull is weaker, because the context determines the masked target [3, 13].

Rotary embeddings are relative, but our design still leaks absolute position. The illumination token sits at a fixed position s outside the grid, so the logit of a query at cell p to it, $x^\top W_q^\top R(s-p) W_k y$, depends on p even when the query content x is the same at every cell, as it is for our learned queries. Windowed attention [98] with zero-padded border windows and zero-padded convolutions [43, 47] add further channels.

H2. A cross-modal JEPA with partly predictable targets and shared position, product and location channels admits a low-loss solution that encodes these channels and not content, training can converge to it, and absolute-position channels make it easier to reach. Then runs that favor the shortcut should reach a low loss while a linear probe of physical content fails on held-out footprints, and removing the sentinel’s position, hiding the target cells from the context or dropping the location embedding from the targets should delay or prevent the collapse.

3.9. Signals and regularizers that cannot see it

Equation (7) predicts which common health signals stay blind to a position code, and which regularizers reward it.

- *Loss* falls as c_t fades, since whatever cannot be predicted is removed from the target.
- *Effective rank and token cosine gap within a tile* are computed across the positions of one tile. A position code has G^2 distinct tokens there and scores near the maximum on both.
- *VICReg variance*, the per-feature standard deviation over the flattened tokens, is met by $u(i)$ alone.
- *VICReg covariance* on flattened tokens splits into a within-tile part and a between-tile part, $C = C_{\text{within}} + \frac{1}{K} \sum_b (v_b - \bar{v})(v_b - \bar{v})^\top$, where v_b is the mean token of tile b among K tiles. A per-tile content component makes the second term large and correlated across features. A position code makes it vanish. At fixed within-tile structure the penalty is therefore lowest, in expectation, for the shortcut (Appendix B).
- *Pooled SIGReg* [10] on tile means $\bar{m}$ conflicts with the scale of Eq. (4). Layer-normalized tokens give $\|\bar{m}\|^2 = D \bar{c}$, where $\bar{c}$ is the mean cosine between tokens of a tile. Matching $\mathcal{N}(0, I)$ needs $\bar{c} \approx 1$, that is, identical tokens within every tile, so the term can only push toward smoother tiles.
- *The VICReg variance hinge on predictions* asks for a per-feature standard deviation of at least γ . A calibrated prediction $\mathbb{E}[\bar{z} \mid \text{source}]$ has a standard deviation of about $\sqrt{R^2}$, where R^2 is the share of target variance the source explains. For weak pairs R^2 is small. The cheapest way to add the missing spread without raising the loss is structure the target also has and the predictor can produce without

reading the source: u , m_t and l .

H3. The VICReg covariance penalty, a variance hinge on the predictions and pooled SIGReg do not prevent position-code collapse, and the first two push toward it. Then raising the covariance weight or keeping the hinge on the predictions should lower the cross-sample gap of Sec. 3.10, and pooled SIGReg should settle at a plateau set by $\bar{c}$.

3.10. Cross-sample monitors

A position code looks the same in every tile, so it can only be seen by comparing tiles. Let $\tilde{z} = z - \frac{1}{G^2} \sum_i z_i$ denote a token grid with its tile mean removed. For prediction $\hat{z}_b$ and target $\tilde{z}_b$ of sample b in a micro-batch, and $\pi(b)$ another sample with the same target product, we define

$$\Delta = \mathbb{E}_b [\cos(\tilde{z}_b, \tilde{z}_b) - \cos(\tilde{z}_b, \tilde{z}_{\pi(b)})], \quad (8)$$

$$S = 1 - \Delta / \mathbb{E}_b [\cos(\tilde{z}_b, \tilde{z}_b)], \quad (9)$$

where $\cos$ is averaged over the G^2 cells and $\tilde{z}$ is layer-normalized. Δ measures how much better a prediction matches its own footprint than another footprint's, and S is the share of the agreement that another footprint's target would have received anyway. Removing the tile mean cancels m_t and l , which are constant within a tile, so neither product identity nor location can inflate Δ . Comparing against a target of the same product also keeps product-specific texture statistics out of the gap. A position code gives $\Delta \rightarrow 0$ and $S \rightarrow 1$. In a dimensional collapse all targets become alike, so the agreement grows faster than the gap and S should rise even if Δ does not fall. The cost is one extra cosine per target and no extra forward pass.

The monitors cannot see predictions that stay specific to the footprint but lose detail within the tile, so we also log the within-tile token cosine gap. At micro-batch size one, $\pi(b)$ must come from another device or step. Being measured against each run's own EMA target, their absolute values compare only runs whose targets are alike.

H4. Δ and S flag both collapse modes before within-sample statistics do. Then S should separate runs that probes label as collapsed from healthy ones, and cross its alarm threshold before the content probe fails.

3.11. Ground-aligned tokens and dense losses

The stems of Sec. 3.3 and the context loss of Eq. (5) are design choices, and we treat them as claims to test. A shared canvas discards detail in the fine products before any stem sees it. Native stems keep that detail, but without a floor they give the coarsest products one pixel per token and a rank-one target. The cross-modal loss constrains the context tokens only through what the predictor reads from them, while $\mathcal{L}_{\text{ctx}}$ supervises each visible token at its own cell.

H5. Native-resolution, ground-aligned stems with a kernel floor and centered convolution stages improve dense cross-modal alignment and dense downstream tasks over a shared resampled canvas, and so does the distance-weighted context loss. The floor should matter most for coarse products, whose targets are rank one without it. Centering should remove a fixed offset of about 0.4 cells (Eq. (2)) between tokens and the ground they stand for.

4. Experiments and results

Table 2 maps each hypothesis of Sec. 3 to the experiment that decides it. The runs are in progress. For each experiment we give the design, the outcome that would support or refute its hypothesis, and the table we will report.

4.1. Setup

Pretraining. The encoder has 217M parameters (104M active per token), the predictor 116M, and the stems 1.7M (linear) to 62M (convolutional) over all 38 product slots. We use AdamW [51] (weight decay 0.05, clipping at 1), bf16, warmup over 10% of the steps with cosine decay, and flash attention [26] at $G \geq 64$. Ablations run at $G=32$ on the seven products of Tab. 1 for 50k steps, at an effective batch of 768 footprint rows, with three seeds per arm. The main model runs at $G=64$ with fixed-GSD, centered convolutional stems. **[TODO: final main-model recipe: steps, batch, learning rate, floor]**

Monitoring. Every 2.5k steps we evaluate a fixed, seeded validation subset with fixed block masks and log the loss, the within-tile rank and token cosine gap, the VICReg terms, Δ and S (Eqs. (8) and (9)), and a variant of the latter two without tile centering or same-product partners.

Collapse labels. Testing H2–H4 needs labels that do not come from the monitors. We use two linear probes on frozen EMA target tokens, fitted on training footprints and scored on held-out ones. The *content probe* regresses, at every cell, physical values from co-registered maps of other products, such as elevation and slope for a WAC token or 643 nm albedo for a DTM token. The *position probe* regresses the cell coordinates. Position is decodable in healthy JEPAs too [93], so it is informative only next to the content probe. A checkpoint is labeled a position code when its content R^2 falls below 10% of the run's maximum while its position R^2 stays above 0.9. It is labeled dimensionally collapsed when the effective rank of tile-mean embeddings over 1024 validation tiles falls below 10% of the run's maximum. **[TODO: confirm thresholds on a pilot]**

4.2. Downstream tasks and baselines

Crater detection. We detect Robbins craters [70] on WAC tiles at 100m (1k and 10k training tiles) and hand-

Table 2. Hypotheses, the outcome that would support each, and the experiment that decides it.

	Claim	Supported if	Test
H1	Cross-modal prediction under a given sun encodes the terrain behind appearance	frozen features beat a same-product JEPA on terrain tasks; shadow AUC drops when the sun is swapped or removed	E1
H2	A position-code attractor exists; absolute-position channels speed it up	some arms reach low loss with a failing content probe; removing the channels delays or prevents it	E2
H3	VICReg covariance and a hinge on predictions push toward it; pooled SIGReg cannot stop it	Δ falls with w_c and with the hinge; SIGReg plateaus	E3
H4	Tile-centered Δ and S detect both collapse modes early	S separates probe-labeled runs and fires before within-sample signals	E4
H5	Ground-aligned stems, floor, centering and the dense loss improve dense alignment	per-pair Δ and dense tasks improve, most for coarse products	E5

508 labeled craters on NAC images at about 1 m, with an FCOS
509 head [85] on a feature pyramid from four encoder depths.
510 We report COCO mAP and mAP₅₀ with a fully frozen back-
511 bone and with fine-tuned stems, and also train with the DTM
512 as a second input or with a DTM that the model predicts
513 from the WAC image.

514 **Irregular mare patches.** We segment IMPs [19] on NAC
515 images from SomBench [60] and report IoU.

516 **Shape and albedo from shading.** On held-out footprints
517 observed under at least six suns, probes read relief and
518 albedo from the predictor features computed from a single
519 WAC image. We report correlation with the plane-detrended
520 DTM and the 643 nm mosaic, invariance across observa-
521 tions of a footprint, and top-1 retrieval of the footprint from
522 its predicted relief.

523 **Baselines.** We compare with the NASA-IBM lunar foun-
524 dation model [31] under its published protocol, a generic
525 self-supervised ViT-B (DINOv3 [76]) and the controls of
526 E1. We will add LunarFM [34] and Moonstone [64] where
527 weights are available. [TODO: confirm availability]

528 4.3. E1: What the objective learns

529 We compare LeLunar with two controls of identical archi-
530 tecture and budget: a same-product JEPA, whose targets are
531 masked cells of the source product itself as in I-JEPA, and
532 LeLunar without illumination conditioning. A randomly
533 initialized encoder gives the floor. All are evaluated on the
534 tasks of Sec. 4.2. For the sun-swap test we take held-out
535 footprints with at least six WAC observations. We decode
536 WAC shadows from the predictor under the observation’s
537 true sun, the sun of another observation of the same foot-
538 print, and no sun, and score shadow localization by AUC on
539 high-incidence observations.

540 H1 is supported if LeLunar beats the same-product JEPA
541 on terrain-dependent tasks at matched compute (Tab. 3), and
542 if removing the sun from the predictor costs most on the
543 shadow test. If the same-product control matches LeLunar,

the cross-modal target adds nothing that same-image predic-
tion does not already give. [TODO: fill after E1]

544 4.4. E2, E3: Position code and regularizers 546

547 **E2 (H2).** Starting from the baseline recipe, each arm adds
548 or removes one absolute-position channel: (a) windowed
549 long-short attention [98] in the encoder; (b) illumination
550 through AdaLN only, without the sentinel token; (c) target
551 cells hidden from the context, so that the loss is computed
552 only where the source is masked, as in I-JEPA; (d) no loca-
553 tion embedding on the target tokens.

554 **E3 (H3).** We sweep the covariance weight over
555 $\{0, 0.005, 0.05\}$ and set the variance hinge on the pre-
556 dictions to $\gamma \in \{1, 0.3\}$ or off, keeping the hinge on
557 the context. We compare pooled SIGReg (weight 0.003)
558 with SIGReg on a small projector of the pooled context
559 embedding, the setting LeJEPA was designed for [10].

560 H2 is supported if at least one arm of Tab. 4 reaches a
561 low loss with a failing content probe, and if arms without
562 absolute-position channels collapse later or not at all. H3
563 is supported if Δ falls as the covariance weight grows and
564 rises when the hinge on the predictions is removed. [TODO:
565 fill after E2, E3; add curves of a healthy and a collapsed
566 arm]

567 4.5. E4: Do the monitors work?

568 On all checkpoints of E2, E3 and E5 we treat each monitor as
569 a detector of the probe labels. We report the area under the
570 ROC curve and the lead time, the number of steps between
571 the monitor’s first alarm and the onset of the label.

572 H4 is supported if the tile-centered S has the highest
573 AUROC and a positive lead time for both collapse modes
574 (Tab. 5). If the uncentered variant does as well, centering is
575 unnecessary; if within-sample signals do, the cross-sample
576 comparison is. [TODO: fill after E4]

577 4.6. E5: Tokens and dense losses

578 Stems form a 2×2 design (shared canvas or fixed-GSD, lin-
579 ear or convolutional). On top of fixed-GSD convolutional
580 stems we vary centering ($k=3$ or $k=s+2$), the kernel floor

Table 3. Downstream results with frozen encoders (E1). Crater columns: mAP₅₀ on Robbins 1k and 10k (WAC, 100 m) and on NAC (≈ 1 m). IMP: IoU. Relief and albedo: correlation of linear probes on held-out footprints. Shadow AUC under the true sun, another observation’s sun and no sun. Entries will be filled when the runs finish.

Encoder	Craters (mAP ₅₀)			IMP IoU	Shape from shading		Shadow AUC		
	1k	10k	NAC		relief	albedo	true	other	none
Random initialization	–	–	–	–	–	–	–	–	–
DINOv3 ViT-B [76]	–	–	–	–	–	–	–	–	–
NASA-IBM LFM [31]	–	–	–	–	–	–	–	–	–
Same-product JEPA	–	–	–	–	–	–	–	–	–
LeLunar without sun	–	–	–	–	–	–	–	–	–
LeLunar	–	–	–	–	–	–	–	–	–

Table 4. Pretraining signals at the end of each arm of E2 and E3 (mean over three seeds). $\mathcal{L}_{\text{cm}}$: validation loss. Rank: effective rank within a tile. $\mathcal{C}$: VICReg covariance. R_c^2 : content probe.

Arm	$\mathcal{L}_{\text{cm}}$	Rank	$\mathcal{C}$	R_c^2	Δ	S
Baseline	–	–	–	–	–	–
+ long-short attention	–	–	–	–	–	–
Sun via AdaLN only	–	–	–	–	–	–
Target cells hidden from context	–	–	–	–	–	–
No location on targets	–	–	–	–	–	–
$w_c=0 / 0.05$	–	–	–	–	–	–
Hinge on $\hat{z}$: $\gamma=0.3$ / off	–	–	–	–	–	–
Pooled SIGReg / projector SIGReg	–	–	–	–	–	–

Table 5. Monitors as collapse detectors over all checkpoints of E2, E3 and E5: AUROC against the probe labels and lead time (k steps before label onset) for position-code (P) and dimensional (D) collapse.

Monitor	AUROC		Lead time	
	P	D	P	D
Validation loss	–	–	–	–
Within-tile rank	–	–	–	–
Within-tile token cosine gap	–	–	–	–
VICReg covariance	–	–	–	–
S , uncentered	–	–	–	–
S , tile-centered (ours)	–	–	–	–

($k_{\min} \in \{1, 4\}$, with the 12 products of Tab. 1 at $G=64$), the dense context loss ($\lambda_{\text{ctx}} \in \{0, 0.5\}$, $\tau \in \{1, 2, 4\}$) and the grid ($G \in \{32, 64, 128\}$, one seed at 128).

H5 is supported if the kernel floor helps most on pairs with coarse products, and if centering and the dense loss help on dense tasks (Tab. 6). If the shared canvas matches the fixed-GSD stems, their added complexity is not worth it. **[TODO: fill after E5]**

5. Limitations and conclusion

Limitations. The hypotheses rest on an informal decomposition (Eq. (7)) and on expectations over random directions, not on a proof that training reaches the position code.

Table 6. Stem and dense-loss variants (E5). Δ over all product pairs and over pairs with a coarse product (roughness, TiO₂, WAC UV), content probe R_c^2 , crater mAP₅₀ (Robbins 1k) and relief correlation.

Variant	Δ_{all}	Δ_{coarse}	R_c^2	Craters	Relief
Shared canvas, linear	–	–	–	–	–
Shared canvas, conv	–	–	–	–	–
Fixed-GSD, linear	–	–	–	–	–
Fixed-GSD, conv ($k=3$)	–	–	–	–	–
Fixed-GSD, conv, centered	–	–	–	–	–
+ $k_{\min}=4$	–	–	–	–	–
+ dense loss ($\tau=2$)	–	–	–	–	–

The probe labels of Sec. 4.1 are themselves a choice: a content probe can miss content that is present but not linearly decodable. Even tile-centered, the monitors cannot see a loss of detail within the tile, and they compare only runs whose targets are alike. Polar regions and the 1 m NAC scale, where much of the scientific demand lies, are outside the planned runs.

Conclusion. We proposed predicting one lunar instrument from another in latent space, with a predictor told the sun, as a pretraining task for co-registered planetary data. We argued why it should teach the static maps what shapes the appearance of the surface, and why the same design invites a position code that the loss, within-sample statistics and VICReg read as progress. The experiments of Sec. 4 are built to decide these claims. The tile-centered cross-sample monitors are cheap enough to log in any JEPA whose targets are only partly predictable from the context. **[TODO: revise with results]**

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LeLunar: Learning the Moon by Predicting One Instrument from Another

Supplementary Material

A. Receptive-field offset of strided conv stems

Consider one stage with stride s , kernel k and padding q on a one-dimensional input of length sn (pixels $0, \dots, sn-1$). Output o reads input pixels $so-q, \dots, so-q+k-1$, so its receptive field is centered at $so-q + (k-1)/2$. The cell that output o stands for spans pixels $so, \dots, so+s-1$, with center $so + (s-1)/2$. The offset of one stage is therefore

$$\delta_1 = \frac{k-1}{2} - q - \frac{s-1}{2}. \quad (10)$$

With $k=3$ and $q=1$, $\delta_1 = -(s-1)/2$ input pixels. Stacking stages with strides $s_1, \dots, s_n$ and total stride $P = \prod_j s_j$ gives, in native pixels,

$$\delta = -\sum_j \left(\prod_{i < j} s_i \right) \frac{s_j - 1}{2} = -\frac{P-1}{2}, \quad (11)$$

because the sum telescopes. In units of one token cell (P pixels) this is Eq. (2). With $k=s+2$ and $q=1$, $\delta_1=0$ for every s , the receptive field extends one pixel beyond the cell on each side, and the output length is unchanged. At $s=3$ a 3×3 kernel reads pixels $3o-1, \dots, 3o+1$, so pixel $3n-1$ is never read, and two such stages leave the last four of 576 columns of a WAC tile unused. Stride-2 stages read every pixel at either kernel size. Table 7 lists the offsets for the stems used at $G=64$.

Table 7. Token offset (fraction of a cell, negative is up-left) and unread border columns for the convolutional stems at $G=64$, from the formulas above.

p (strides)	kernel 3	kernel $s+2$	unread cols
4 (2, 2)	-0.375	0	0
9 (3, 3)	-0.444	0	4
16 (2, 2, 2, 2)	-0.469	0	0

B. Derivations for Sec. 3.9

Covariance of flattened tokens. Let $z_{b,i} \in \mathbb{R}^D$ be the tokens of K tiles ($b = 1..K, i = 1..N$), v_b the mean of tile b and $\bar{v}$ the mean over tiles. The covariance over all KN tokens is

$$C = C_{\text{within}} + C_{\text{between}}, \quad (12)$$

$$C_{\text{within}} = \frac{1}{KN} \sum_{b,i} (z_{b,i} - v_b)(z_{b,i} - v_b)^\top, \quad (13)$$

$$C_{\text{between}} = \frac{1}{K} \sum_b (v_b - \bar{v})(v_b - \bar{v})^\top. \quad (14)$$

VICReg’s penalty is $\mathcal{C} = \frac{1}{D} \sum_{j \neq k} C_{jk}^2$. For a position code, $z_{b,i} = u(i)$ in every tile, so $v_b = \bar{v}$ and $C_{\text{between}} = 0$. If instead the tiles carry content with a per-tile component, C_{between} is a sum of K outer products whose off-diagonal entries are of the order of v , the between-tile variance per feature. When the directions of $v_b - \bar{v}$ are independent of C_{within} , the cross term between the off-diagonal parts of the two matrices vanishes in expectation, and

$$\mathbb{E}[\mathcal{C}] = \mathcal{C}(C_{\text{within}}) + \mathbb{E}[\mathcal{C}(C_{\text{between}})] \geq \mathcal{C}(C_{\text{within}}). \quad (15)$$

At fixed within-tile structure the penalty is therefore smallest, in expectation, when the tiles do not differ, which is the position code. The per-rank batches used in practice hold only a few tiles (K is the number of target grids per device), so C_{between} is low-rank and its off-diagonal energy is large relative to v .

Pooled SIGReg. Let $t_1, \dots, t_N$ be the layer-normalized tokens of a tile, so that $\|t_i\|^2 = D$, and let $\bar{m} = \frac{1}{N} \sum_i t_i$. Then

$$\|\bar{m}\|^2 = \frac{1}{N^2} \sum_{i,j} t_i^\top t_j = D \bar{c}, \quad \bar{c} = \frac{1}{N^2} \sum_{i,j} \cos(t_i, t_j). \quad (16)$$

Under $\mathcal{N}(0, I_D)$, $\mathbb{E} \|\bar{m}\|^2 = D$, so a pooled prediction matches the SIGReg target only when $\bar{c} \approx 1$, that is, when all tokens of the tile are nearly identical. A prediction pinned to layer-normalized, spatially varying targets cannot satisfy it. What remains is a pressure toward smoother tiles that does not depend on whether the tile mean carries content.

Variance hinge on predictions. For a squared-type loss the optimal prediction is $\hat{z} = \mathbb{E}[\bar{z} \mid \text{source}]$, and the law of total variance gives, per feature d , $\text{Var}(\hat{z}_d) = \text{Var}(\bar{z}_d) - \mathbb{E}[\text{Var}(\bar{z}_d \mid \text{source})] = R_d^2 \text{Var}(\bar{z}_d)$. With layer-normalized targets $\text{Var}(\bar{z}_d) \approx 1$, so the calibrated standard deviation is about $\sqrt{R_d^2}$. A hinge at $\gamma=1$ can be met only by adding variance beyond the calibrated part. Variance that the source does not explain raises the loss unless the target shares it. The components that every target shares, and that the predictor can produce without the source, are u , m_t and l of Eq. (7).

C. Planned runs

Shared recipe. Unless stated otherwise, runs use a ViT-B encoder whose last four blocks are MoE (8 experts, top-2), a 12-layer predictor of width 768, four supervision levels, a visible square block covering 70–80% of the grid and $K=2$

targets per row. The VICReg variance weight is 1 on context and predictions, the covariance weight 0.005, SIGReg is off, and the EMA decay is a constant 0.996. We use AdamW with weight decay 0.05 and clipping at 1, warmup over 10% of the schedule and cosine decay to 10^{-6} . Ablation arms run at $G=32$ with seven products for 50k steps and three seeds. **[TODO: list every arm with its identifier once launched]**

D. Downstream protocols

Crater detection. Robbins 1k has 800/100/100 WAC tiles of 512 px at 100 m, Robbins 10k has 8000/1000/1000 tiles with a wider range of illumination, and the NAC set has 289/63/56 hand-labeled tiles of 256 px at about 1 m. Detectors read the frozen EMA encoder at its four supervision depths through a trainable three-level feature pyramid with FCOS heads, trained on multi-scale native crops. We report two settings: fully frozen, and with the per-product stems fine-tuned at learning rate 10^{-4} . We score all boxes down to 2 px with at most 300 detections. For the comparison with the NASA-IBM model we also follow its protocol: boxes with a shortest side of at least 10 px, tiles without craters dropped, at most 500 detections, and a Faster R-CNN head family. The predicted DTM comes from a DPT head trained on the model's elevation-product predictor features, given the WAC image.

Shape and albedo from shading. Footprints with at least six observations are split, with a fixed seed, into a fitting and an evaluation set. Outputs are 128 px (469 m per pixel). The relief target is the plane-detrended 60 m DTM. Albedo is scored against the 643 nm mosaic and, separately, against a multi-sun pseudo-albedo. Linear probes read frozen predictor features. Shadow AUC uses the darkest-quantile pixels of the WAC image as labels, on high-incidence observations only, where dark pixels are mostly cast shadow. **[TODO: IMP segmentation protocol]**